

Engineering Mechanics and Strength of Materials

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

3123

Credits

4

Periods

5/week

Course Details

Title: Engineering Mechanics and Strength of Materials

Department: Common

Revision: REV_Engineering Mechanics and Strength of Materials

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 3123 — Engineering Mechanics and Strength of Materials**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ു൬൬൬൬൬൬ ു൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ English technical terms- ു൬൬൬൬൬൬൬൬൬. ു൬ ു൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬, formula, diagram, units, definitions ു൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

TABLE OF CONTENTS

Navigate handbook

Objectives & Outcomes

Module 1

Module 2

Module 3

Module 4

Model Question Paper

Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Engineering Mechanics and Strength of Materials.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Engineering Mechanics and Strength of Materials.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഐക്യ പ്രതിബദ്ധതയുടെ അടിസ്ഥാനപരമായ ആശയം പരിചയപ്പെടുത്തുക. ഐക്യ പ്രതിബദ്ധതയുടെ അടിസ്ഥാനപരമായ ആശയം പരിചയപ്പെടുത്തുക.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ പ്രയോജനപ്പെടുത്തുന്നതിനുള്ള സാധനങ്ങൾ ഉപയോഗിക്കുക. ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ പ്രയോജനപ്പെടുത്തുന്നതിനുള്ള സാധനങ്ങൾ ഉപയോഗിക്കുക.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: സങ്കീർണ്ണമായ സാങ്കേതികവിദ്യകൾ സംയോജിപ്പിക്കുന്നതിനുള്ള സാധനങ്ങൾ ഉപയോഗിക്കുക. സങ്കീർണ്ണമായ സാങ്കേതികവിദ്യകൾ സംയോജിപ്പിക്കുന്നതിനുള്ള സാധനങ്ങൾ ഉപയോഗിക്കുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഐക്യം നമ്മുടെ ഏകതയാണ്. ഐക്യം നമ്മുടെ ഏകതയാണ്. ഐക്യം നമ്മുടെ ഏകതയാണ്.

MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Engineering Mechanics and Strength of Materials.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.